

Andrew L. Ikemoto

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EDUCATION

Rutgers University–New Brunswick – *School of Engineering*

Expected May 2028

- B.S. in Mechanical Engineering, Minor in Computer Science

TECHNICAL SKILLS

Fabrication & Assembly: Shop tool operation (Band saw, Miter saw, Drill press, Jigsaw, Dremel), Tapping/Die

Analysis & Simulation: ANSYS, Ftool

Software & Tools: SolidWorks, AutoCAD, Onshape, MATLAB, Python, Arduino

Manufacturing & Prototyping: 3D printing (PrusaSlicer, Bambu Studio), Machining, CNC

EXPERIENCE

Rutgers University Marine Field Station (RUMFS)

May 2025 – Aug. 2025

Design Research Intern

- Documented and communicated the **design** and **capabilities** of advanced **ocean systems**, including **autonomous surface and underwater vehicles (ASVs/AUVs)** and **offshore energy platforms** for researchers and entrepreneurs.
- Communicated with 4 marine engineers and researchers at the **Rutgers U** and **Stockton U** Marine Field Stations.
- Analyzed **marine vehicle architecture and engineering tradeoffs**, including modular hull configurations, articulated hulls, propulsion systems, equipment mounting, transport mechanisms, and satellite communications.
- Examined OPT's mechanical product development workflow from **SolidWorks CAD** and **internal design review** through **external manufacturing**, including considerations of design integrity, cost, sourcing, and fabrication.

J-Star Research Inc.

June 2024 – Sept. 2024

R&D Intern

- Conducted R&D on antibody–drug conjugation systems; used analytical instruments such as NMR, MS, and HPLC.
- Followed detailed lab protocols and maintained documentation of experimental procedures and compound analyses.

PROJECTS & LEADERSHIP

Rutgers ASME — *eHPVC Team Lead*

Aug. 2025 – Present

- Led a **30-member engineering team** in building the design, chassis, handling, and ergonomics of an **electric vehicle**.
- Oversaw manufacturing, communications, and logistics of the team to compete in the Spring.
- Represented Rutgers' 400-member ASME chapter on the Governing Council, advocating for student opportunities.
- Initiated Rutgers' participation in **e-HPVC (Electric & Human-Powered Vehicle Competition)**.

A4A (Accessible 4 All) — *Project Lead x2*

Feb. 2024 – Present

- Led a **25-member team** in the redesign of an **adaptive tricycle**, overseeing **CAD**, **fabrication**, and **testing**.
- Designed and fabricated using **SolidWorks**, **8020**, **PVC**, and **3D printing** to meet safety and usability requirements.
- Previously redesigned an RC vehicle with custom structural supports, applying prototyping and fabrication techniques.

Rutgers Formula Racing — *Chassis Subteam Member*

Mar. 2026 – Present

- Contributed to the mechanical design of the chassis for the **Formula SAE** competition.
- Building on prior **FRC robotics experience** in **CAD** and collaborative engineering design.

Innovation, Design, & Entrepreneurship Academy — *Fellow Leader*

Sept. 2024 – May 2026

- Mentoring a **cohort of 20 students** in product design, innovation, and entrepreneurship through hands-on projects.
- Pitched a **drone-based search-and-rescue system** to industry professionals, **winning a campus pitch competition**.

ACADEMIC HONORS & AWARDS

- *Break Through Tech Scholar* (AI+ML at Cornell Tech) 2026
- *Road to Silicon Valley (RSVP)*, Cohort 7 Member 2026
- *Richard N. Weeks Scholarship Recipient* 2025
- *AI Inspirit Scholar* (project-based AI+ML program) 2023
- *Hispanic Scholarship Fund Scholar* 2023 – Present
- *American Society of Mechanical Engineers (ASME)* – Member 2025 – Present